

BEN GURION UNIVERSITY OF THE NEGEV

Department of Industrial Engineering & Management

Final Project - Selected Topics in Statistics

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Table of Contents

1. Abstract 3

2. Introduction 4

3. Research Question 5

4. Method 5

4.1 Data 5

4.2 Variables 6

4.3 Pre-Processing 13

4.3.1 Missing values 13

4.3.2 Duplicate records 13

4.3.3 Data proportion 13

4.3.4 Handling outliers 14

4.3.5 Pearson correlation matrix 16

4.4 Detecting Multicollinearity with VIF 16

4.5 Feature Re-presentation 17

5. Models 17

5.1 Binary Logistic Regression 17

5.2 Decision tree (DT) 22

5.2.1 Model preparation 22

5.2.2 Hyperparameter Tuning 23

5.2.3 Model Results 24

5.3 Random forest (RF) 25

5.3.1 Hyperparameter Tuning 25

5.3.2 Model Results 25

5.4 Artificial Neural Network (ANN) 26

5.4.1 Model preparation 26

5.4.2 Hyperparameter Tuning 26

6. Discussion and Results 28

7. References..... 29

8. Appendixes..... 30

1. Abstract

Cardiovascular diseases are a major cause of mortality worldwide, particularly through myocardial infarctions and heart failures. Machine learning applied to electronic medical records can uncover patterns and correlations that aid in predicting patient survival. In this project, we analyze data from 299 heart failure patients in 2015. Using machine learning classifiers, we predict survival and rank features by importance. Time, serum creatinine and ejection fraction emerge as the most crucial features, enabling highly accurate survival predictions when used alone. This finding could impact clinical practice, suggesting that focusing on these three features could improve survival prediction in heart failure patients.

2. Introduction

Cardiovascular diseases (CVDs) encompass various heart and blood vessel disorders, including coronary heart disease, cerebrovascular diseases, and heart failure (HF). Globally, CVDs contribute to approximately 17 million annual deaths, marking an increasing trend in the United Kingdom after 50 years [1]. Heart failure, a condition where the heart struggles to pump sufficient blood, is often linked to diabetes, high blood pressure, or other heart-related issues [2]. Clinically, heart failure is categorized based on ejection fraction, representing the proportion of blood ejected during a heart contraction, with HFrEF (reduced ejection fraction) having an ejection fraction below 40%, and HFpEF (preserved ejection fraction) characterized by normal contraction but impaired relaxation [3].

Clinicians assess disease progression quantitatively using the New York Heart Association (NYHA) functional classification. This classification has four classes, from no symptoms during ordinary activities (Class I) to symptoms occurring at rest with any physical activity (Class IV). Despite it is common use, there is no uniform method for evaluating the NYHA score [4].

Predicting heart failure remains a challenge for medical professionals, with clinical practice struggling to achieve high accuracy [5]. Medical health records (EHRs), emerge as valuable resources for uncovering hidden correlations in patients' data. They not only serve research purposes but also offer insights for clinical practice and dispelling traditional myths about risk factors [6].

Machine learning in medical records effectively predicts heart failure patient survival [7] and identifies crucial risk factors [8]. It provides valuable insights for clinical prediction and feature ranking. Recent studies on deep learning and meta-analysis enhance performance compared to human specialists, albeit with slightly lower accuracy (0.75 versus 0.59) [9].

Modeling survival in heart failure and cardiovascular diseases faces challenges in achieving high prediction accuracy and understanding driving factors. Many existing models offer only modest accuracy, limited interpretability, and lack consensus on predictors' impact [10].

The dataset analyze is from a case study on survival analysis of heart failure patients who were admitted to Institute of Cardiology and Allied hospital Faisalabad-Pakistan during April-December 2015.

In this project, our primary goal is to predict the survival of patients, emphasizing the identification and ranking of features according to their significance as risk factors. The focus lies on establishing a comprehensive understanding of the relationships between various factors and their impact on patient survival.

3. Research Question

In this project, our objective is to predict the survival of patients who experienced heart failure. We will analyze various risk factors and assign rankings to prioritize them based on their respective levels of importance.

4. Method

4.1 Data

The dataset encompasses medical records from 299 patients diagnosed with heart failure (HF) collected in 2015. The cohort comprises 105 women and 194 men. All patients included in the dataset were aged over 40, exhibited left ventricular systolic dysfunction, and were classified under NYHA (New York Heart Association) class 3 and 4 [11]. HF occurs when the heart is unable to pump enough blood to the body. The clinical community groups it into two types based on ejection fraction value. Ejection fraction value - proportion of the amount of blood the heart pumps out in a single contraction.

HF two types due to left ventricular dysfunction [12]:

1. **Systolic** - the heart can't pump with enough force to push enough blood into circulation. This is also known as heart failure with reduced ejection (40% or less).
2. **Diastolic** - the left ventricle loses its ability to relax normally and properly fill with blood during the resting period because the muscle has become stiff. This is also known as heart failure with mid-range ejection (40%-49%).

All patients had left ventricular systolic dysfunction and experienced HF before.

4.2 Variables

Table 1 Variables - measurement units and interval of each feature in the dataset

Variable	Meaning	Type	Units	Range
Age	Age of the patient	Integer	Years	[40,95]
Anemia	A patient having anemia if hematocrit [1] levels were lower than 36%	Boolean	-	{1 - True, 0 - False}
Diabetes	If the patient has diabetes	Boolean	-	{1 - True, 0 - False}
Creatinine phosphokinase (CPK)	The level of the CPK [2] enzyme in blood	Numeric	mcg/L	[23,7861]
Ejection fraction	Proportion of the amount of blood the heart pumps out in a single contraction	Numeric	Percentage (%)	[14,...,80]
Sex	Gender of the patient	Boolean	-	{1 - Man, 0 - Woman}
Diabetes	If the patient has diabetes	Boolean	-	{1- True, 0 - False}
Platelets	Platelets[3] in the blood Platelets	Numeric	k/mL (thousands per milliliter)	[25.01,...,850]
Serum creatinine	Level of creatinine in the blood. Creatinine - a waste product that comes from the digestion of protein in food and the normal breakdown of muscle tissue. It is removed from the blood through our kidneys	Numeric	mg/dL (milligrams per deciliter)	[0.5,...,9.4]

Variable	Meaning	Type	Units	Range

[1] measure of the volume of red blood cells relative to the total volume of a blood sample

[2] an enzyme (a protein that helps to elicit chemical changes in your body) found in the heart, brain, and skeletal muscles. When muscle tissue is damaged, CPK leaks into the blood. Therefore, high levels of CPK usually indicate some sort of stress or injury to the heart or other muscles

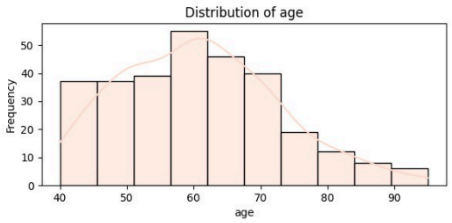
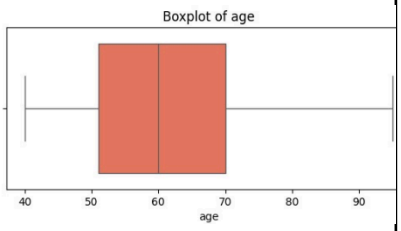
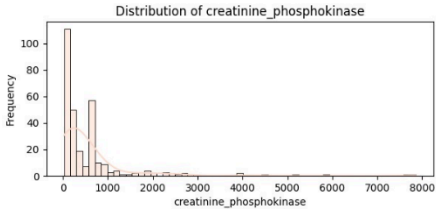
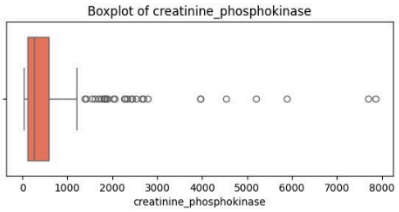
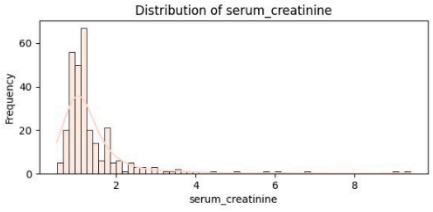
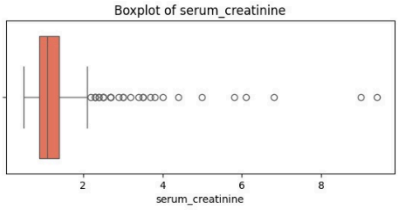
[3] the cells that circulate within our blood and bind together when they recognize damaged blood vessels

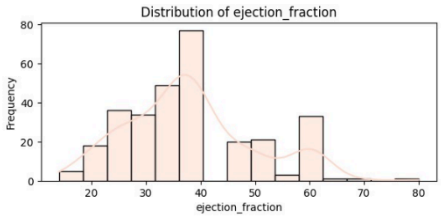
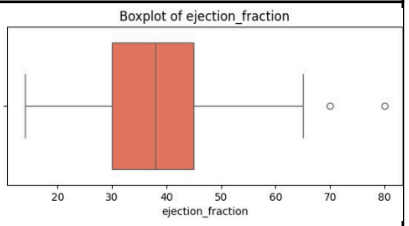
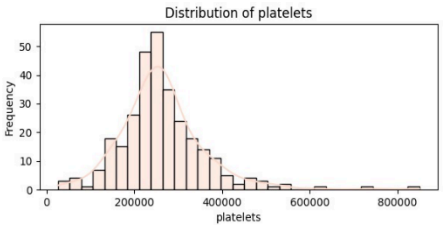
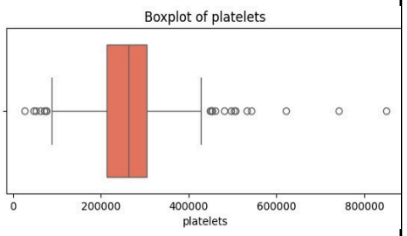
Serum sodium	Level of sodium in the blood. Sodium – mineral that serves for the correct functioning of muscles and nerves. An abnormally low level of sodium might be due to HF	Numeric	MEq/L (milliequivalents per liter)	[114,..,148]]
Smoking	If the patient smokes	Categorical	Boolean	{1- True, 0 - False}
High blood pressure	There is no specific definition of what values of blood pressure considered high	Categorical	Boolean	{1- True, 0 - False
Time	Follow-up period, the time the blood test was taken after HF evet	Numeric	Days	[4,...,285] with average of 130 days
Death event]Target]	If the patient died during the follow-up period	Categorical	Boolean	{1- True, 0 - False

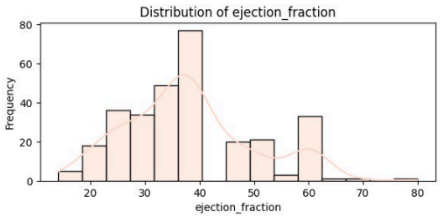
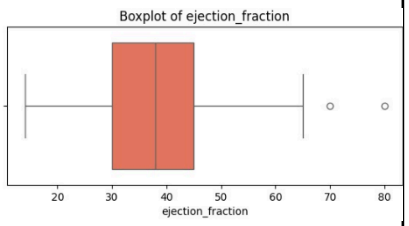
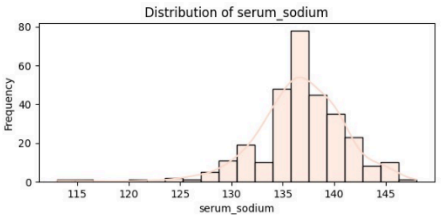
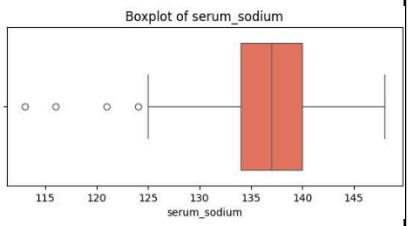
Table 2 Statistical quantitative description of the variables

	count	mean	std	min	25%	50%	75%	max
age	299.00000	60.83389	11.89481	40.00000	51.00000	60.00000	70.00000	95.00000
anaemia	299.00000	0.43144	0.49611	0.00000	0.00000	0.00000	1.00000	1.00000
creatinine_phosphokinase	299.00000	581.83946	970.28788	23.00000	116.50000	250.00000	582.00000	7861.00000
diabetes	299.00000	0.41806	0.49407	0.00000	0.00000	0.00000	1.00000	1.00000
ejection_fraction	299.00000	38.08361	11.83484	14.00000	30.00000	38.00000	45.00000	80.00000
high_blood_pressure	299.00000	0.35117	0.47814	0.00000	0.00000	0.00000	1.00000	1.00000
platelets	299.00000	263358.02926	97804.23687	25100.00000	212500.00000	262000.00000	303500.00000	850000.00000
serum_creatinine	299.00000	1.39388	1.03451	0.50000	0.90000	1.10000	1.40000	9.40000
serum_sodium	299.00000	136.62542	4.41248	113.00000	134.00000	137.00000	140.00000	148.00000
sex	299.00000	0.64883	0.47814	0.00000	0.00000	1.00000	1.00000	1.00000
smoking	299.00000	0.32107	0.46767	0.00000	0.00000	0.00000	1.00000	1.00000
time	299.00000	130.26087	77.61421	4.00000	73.00000	115.00000	203.00000	285.00000
DEATH_EVENT	299.00000	0.32107	0.46767	0.00000	0.00000	0.00000	1.00000	1.00000

Table 3 Statistical description of the numeric variables

Variable	Statistical description
Age	  We can see that age range of the patients is between 40 and 90, and the median age of the patients is 60. Also, from the histogram, it can be concluded that most of the patients are in the age range between 50 and 70, with the age distribution skewed slightly to the right (more people over 60 than under 60).
Creatinine phosphokinase	  The range of values for creatinine phosphokinase among patients varies between 0 and 8000 mcg/L and the average level of this enzyme is approximately 600 mcg/L. It can also be seen that the distribution is skewed to the right and that there are many abnormal values that are significantly higher than most of the values of this variable (we note that normal values in a healthy person range from 0 to 120 micrograms per liter. Therefore, we would like to examine the reason for these many abnormal values and how to deal with them.
Serum creatinine	  The range of serum creatinine values among the patients ranges from 0 to 10, with the median amount is being 1.5 mg/dL (we note that

	the normal amount for healthy men is between 0.7-1.3 mg/dL and for women it is between 0.6-1.1 mg/dL). Also, the distribution is skewed to the right and there are also many abnormal values that are above the average amount, so we would like to further examine the reason for these abnormal values and how to deal with them.
Ejection fraction	  The median percentage of the ejection segment among patients is approximately 40% while the average percentage among healthy people varies between 72%-52%. Also, the distribution is not symmetrical and there are a few individual outliers. We would like to examine the reason for these abnormal values and how to deal with them.
Platelets	  The range of values for platelets in patients ranges from 0 to 800,000, with the median value being around 260,000 k/mL (we note that normal values in healthy people are between 150,000-450,000). Also, the distribution is approximately symmetrical and that the variation in values is relatively low (according to the narrow width of the box), this alongside many unusual values. We would like to check the reason for these abnormal values and how to deal with them.

Variable	Statistical description
Ejection fraction	  The median percentage of the ejection segment among patients is approximately 40% while the average percentage among healthy people varies between 72%-52%. Also, the distribution is not symmetrical and there are a few individual outliers. We would like to examine the reason for these abnormal values and how to deal with them.
Serum sodium	  Sodium values range from 115 to 150 mEq/L, with a median of approximately 137 mEq/L. The distribution is symmetrical, with a narrow box indicating low variation. However, there are outliers that need investigation for their cause and appropriate management.

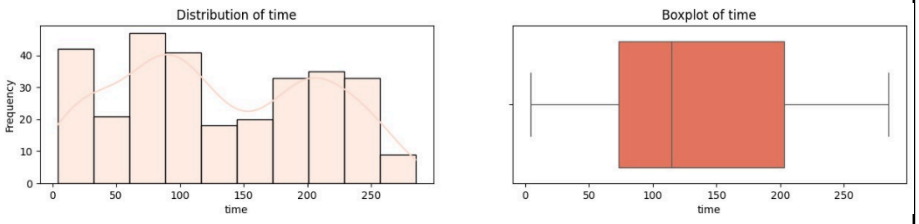
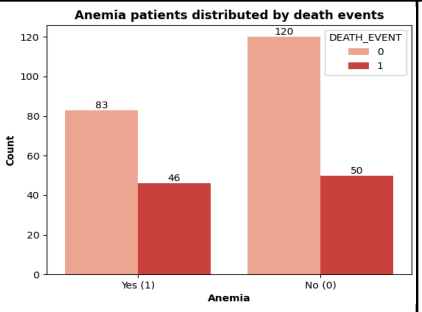
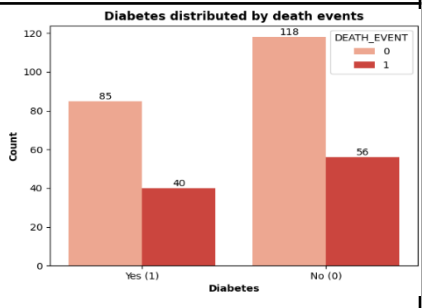
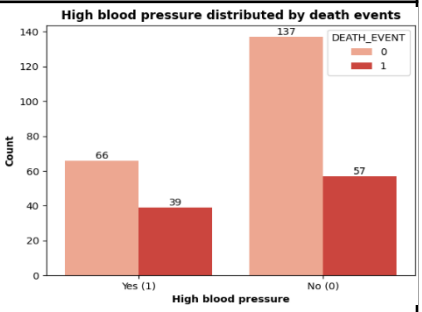
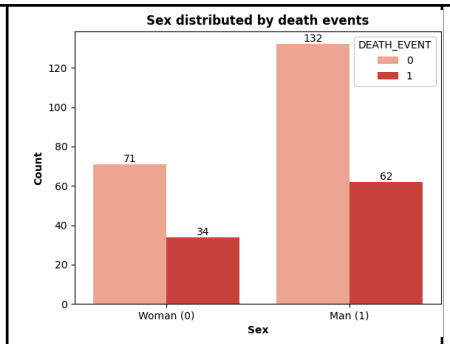
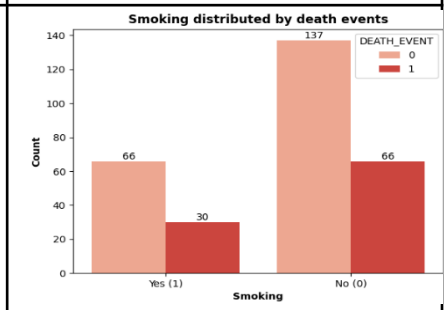
Time	 The range of the time (Follow-up period) varies between 4 to 285 days, which is make sence with the fact the observations were taken during April-December 2015. The median is around 110 and the most patients are in the gap between 7-200 days. There are not outliers records, apperantly because of the time period that was checked.

Table 4 Statistical description of the categorical variables

Variable	Description	Explantion
Anemia		According to the graphs, patients with negative cases of anaemia, diabetes, high_blood_pressure and smoking leads to DEATH_EVENT more than the positive cases of these medical conditions. There are more cases of male population confronting a DEATH_EVENT due to heart failure than female population.
Diabetes		
High blood pressure		

Sex	
Smoking	

4.3 Pre-Processing

4.3.1 Missing values

As part of the data preparation process, we check if there are missing values. There are 299 non-null values in all the attributes thus no missing values. Datatype is either 'float64' or 'int64' - all features are numeric.

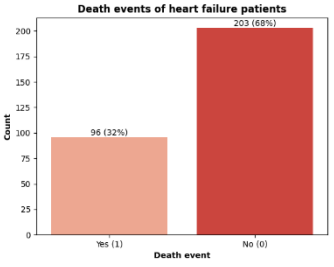
```
>>> df.info()
...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 299 entries, 0 to 298
Data columns (total 13 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   age                       299 non-null    float64
 1   anaemia                   299 non-null    int64
 2   creatinine_phosphokinase  299 non-null    int64
 3   diabetes                  299 non-null    int64
 4   ejection_fraction         299 non-null    int64
 5   high_blood_pressure       299 non-null    int64
 6   platelets                 299 non-null    float64
 7   serum_creatinine          299 non-null    float64
 8   serum_sodium              299 non-null    int64
 9   sex                       299 non-null    int64
 10  smoking                   299 non-null    int64
 11  time                      299 non-null    int64
 12  DEATH_EVENT               299 non-null    int64
dtypes: float64(3), int64(10)
memory usage: 30.5 KB
>>> df.duplicated().sum()
* * *
0
```

4.3.2 Duplicate records

We also checked if there are duplicate records and if there are any, what is their target value. We have chosen to define a duplicate record as a record whose fields are all identical to another record.

4.3.3 Data proportion

During the exploratory data analysis (EDA) stage, we identified an imbalance in the dataset regarding the distribution of samples across categories, between death events and survival cases.



Death events constituted 32% (96 samples) of the dataset, while **survival** cases comprised the remaining 68% (203 samples).

Given the dataset's limited size, we aim to address this imbalance without resorting to **down sampling**, which could result in data loss, or **up sampling**, which may introduce bias. Later, we wish to address the imbalance in the data by choosing proper evaluation metrics:

- **F_2 score**

  An extension of the F1 score, the F2 score allows us to adjust the balance between precision and recall, especially in the presence of imbalanced classes. By varying the value of beta, we can **control the prioritization of precision or recall**. In our case, with $\beta = 2$, we emphasize minimizing False Negatives (capturing as many death events as possible) over minimizing False Positives (ensuring accurate predictions of death events).

  $$F_\beta = \left(1 + \beta^2\right) \bullet \frac{Precision \times Recall}{\left(\beta^2 \times Precision\right) + Recall}$$

- **MCC (Matthews Correlation Coefficient)**

  This metric measures the correlation between predicted and actual binary classifications. It's particularly advantageous for imbalanced datasets as it considers all four outcomes of the binary classification problem.

  $$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{\{(TP+FP)(TP+FN)(TN+FP)(TN+FN)\}}}$$

4.3.4 Handling outliers

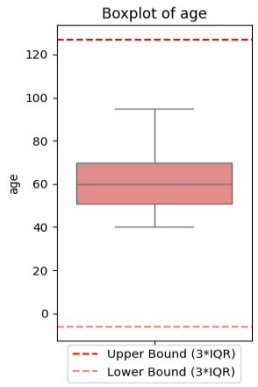
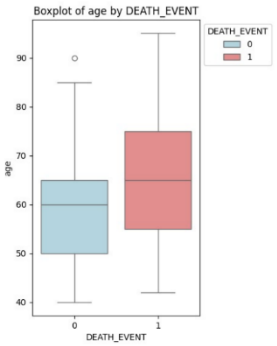
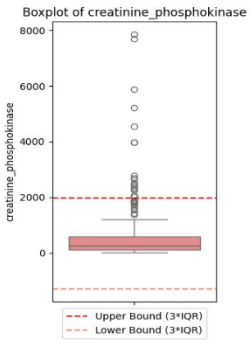
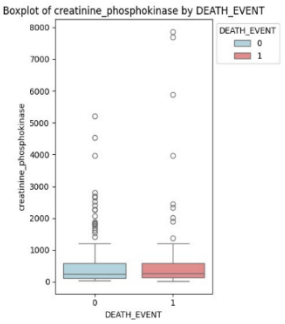
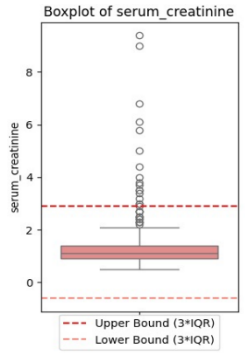
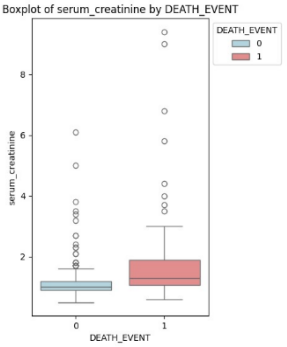
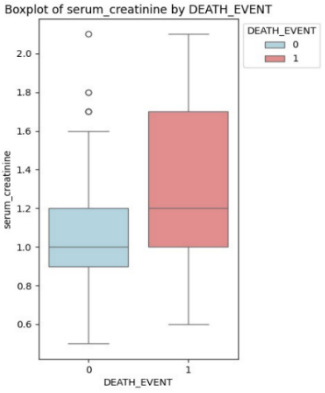
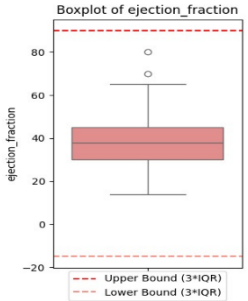
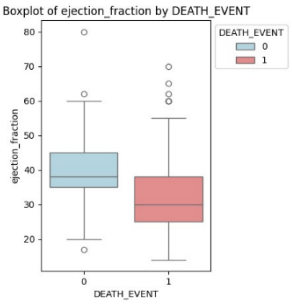
The box plot illustrates a dataset's distribution using key summary statistics: minimum, Q1, median, Q3, and maximum. Outliers, identified using the interquartile range (IQR), fall beyond Q1–1.5 *IQR or Q3 + 1.5* IQR. This criterion, introduced by Tukey, offers a clear method for spotting significantly different data points. The coefficient of 1.5 is adjustable for different applications and assumptions.

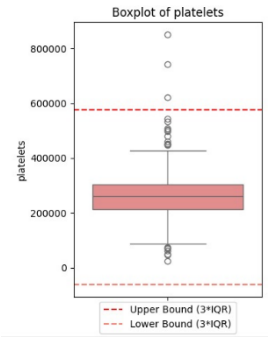
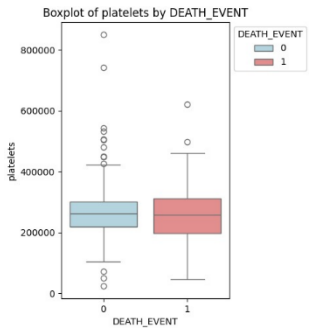
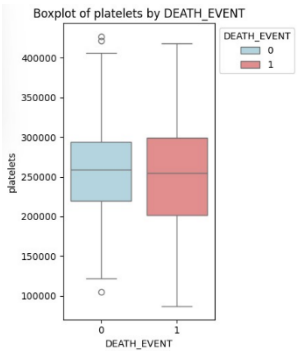
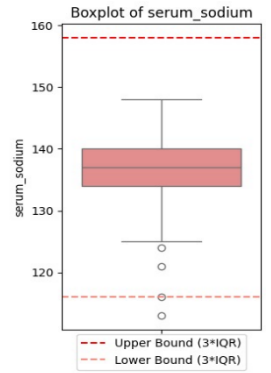
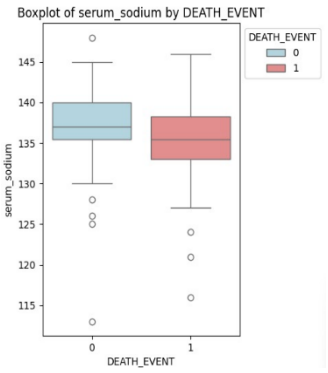
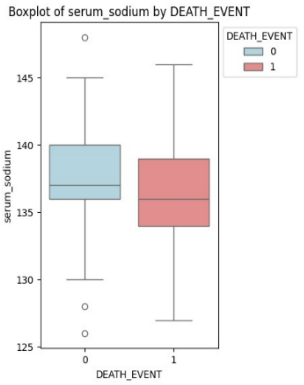
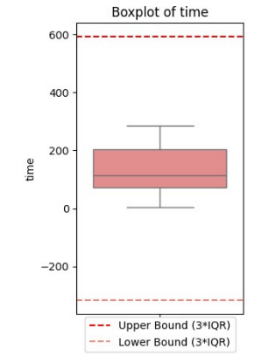
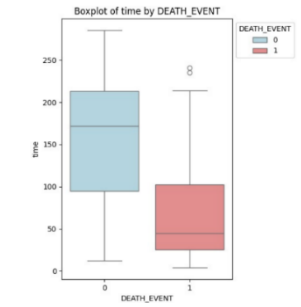
First, we checked for outliers using Tukey's method with coefficient of 1.5. It became apparent that there were numerous outliers in the dataset. To gain a deeper insight into the distribution, we examined the number of outliers in each class of the target variable. We noticed that many outliers were coming from the minority class. Given the imbalance in the data, **we decided to tighten the condition that indicates outliers, and we chose to use coefficient of 3 instead of 1.5,** this is more conservative way that will help us keep the observations and save more data.

Table 5 – Box plot before and after removing outliers

Feature	Box-plot which shows lines of 3*IQR boundaries	Box-plot - Before removing outliers) by classes	Box-plot - after removing outliers according to coefficient=3, by classes

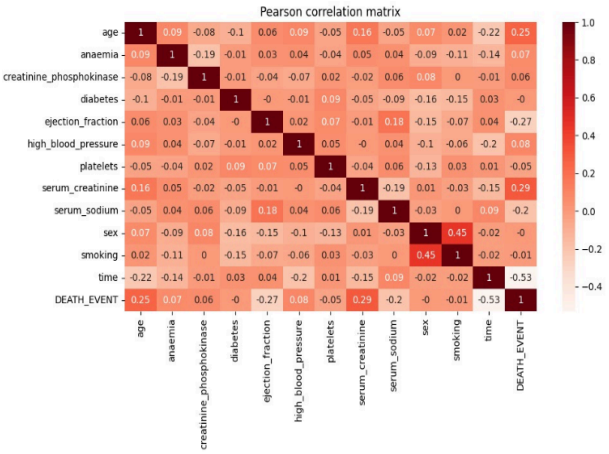
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Age		 No outliers to remove	No Change
Creatinine phosphokinase		 Remove 18 outliers	
Serum creatinine		 Remove 17 outliers	
Ejection fraction		 No outliers to remove	No Change

Platelets		 Remove 3 outliers	
Serum sodium		 Remove 1 outlier	
Time			No Change

4.3.5 Pearson correlation matrix

We use correlation analysis to understand the linear relationship between features and the target variable, but we will not rely on it to handle multicollinearity (for this we will use VIF later).



From the correlation matrix above we found that the correlations among the features [platelets - smoking - sex - diabetes] and the target feature [DEATH_EVENT] are weak correlations, so we assume that these features will be removed during feature selection.

4.4 Detecting Multicollinearity with VIF

Multicollinearity can be detected using various techniques, one such technique being the **Variance Inflation Factor** (**VIF**). In VIF method, we pick each feature and regress it against all the other features. For each regression, the factor is calculated as: $VIF = \frac{1}{1-R^2}$. In our analysis, we adopted a **common rule of thumb:** VIF values greater than or equal to 10 indicate high multicollinearity.

Based on the VIF values (Figure 1), it was observed that the "serum_sodium" and "age" features exhibited high multicollinearity issues. Consequently, we decided to drop the "serum_sodium" feature. Following the removal of "serum_sodium", the VIF analysis was repeated (Figure 2). However, it was still observed that the 'age' feature surpassed the 10 thresholds, indicating continued multicollinearity issues. Therefore, the decision was made to drop the "age" feature as well.

Subsequently, the VIF analysis was conducted again (Figure 3), revealing that the “ejection_fraction” feature did not surpass the VIF=10 threshold, unlike the other features. Therefore, we concluded the analysis at this point.

	feature	VIF
0	age	31.51175
1	anaemia	1.93298
2	creatinine_phosphokinase	2.11919
3	diabetes	1.77788
4	ejection_fraction	12.34396
5	high_blood_pressure	1.72142
6	platelets	10.74602
7	serum_creatinine	10.00027
8	serum_sodium	66.62286
9	sex	3.70514
10	smoking	1.91114
11	time	4.53905

*Figure SEQ Figure * ARABIC 1 - all features*

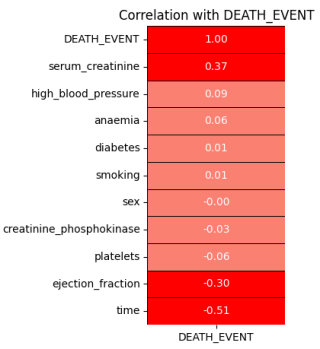
	feature	VIF
0	age	22.38565
1	anaemia	1.85583
2	creatinine_phosphokinase	2.05268
3	diabetes	1.75116
4	ejection_fraction	10.20900
5	high_blood_pressure	1.69274
6	platelets	9.01593
7	serum_creatinine	9.13158
8	sex	3.58848
9	smoking	1.90061
10	time	3.98792

Figure 2 - after removing 'serum sodium'

	feature	VIF
0	anaemia	1.78248
1	creatinine_phosphokinase	2.02547
2	diabetes	1.74882
3	ejection_fraction	8.16708
4	high_blood_pressure	1.63829
5	platelets	8.51231
6	serum_creatinine	6.23510
7	sex	3.41374
8	smoking	1.89846
9	time	3.97635

Figure 3 - after removing 'age'

After we removed the features 'serum sodium' and 'age', we examined again the correlation between the explanatory variables and the explained variable.



Features like 'high_blood pressure', 'anemia', 'creatinine phosphokinase', 'diabetes', 'sex', 'smoking', and 'platelets' have not high or any correlation with DEATH_EVENT. According to this conclusion we assume that as part of the logical steps that we will perform within the various models that we will examine, these elements will not be selected.

4.5 Feature Re-presentation

Normalization Scaling

To mitigate the impact of varying scale values on the correlation with the target binary variable (death event), we employed the normalization scaling technique known as **Min- Max scaling**. This method rescales each feature x such that it falls within the range of [0, 1], thereby reducing potential biases arising from differences in scale.

	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_blood_pressure	platelets	serum_creatinine	sex	smoking	time
0	0	582	0	20	1	265000.00000	1.90000	1	0	4
1	0	7861	0	38	0	263358.03000	1.10000	1	0	6
2	0	146	0	20	0	162000.00000	1.30000	1	1	7
3	1	111	0	20	0	210000.00000	1.90000	1	0	7
4	1	160	1	20	0	327000.00000	2.70000	0	0	8
5	1	47	0	40	1	204000.00000	2.10000	1	1	8

Figure 4 - before scaling
For instance, a noticeable discrepancy exists in the scales of platelets and ejection fraction values.

Top of Form

	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_blood_pressure	platelets	serum_creatinine	sex	smoking	time
0	0.00000	0.07100	0.00000	0.09100	1.00000	0.29100	0.15700	1.00000	0.00000	0.00000
1	0.00000	1.00000	0.00000	0.36400	0.00000	0.28900	0.06700	1.00000	0.00000	0.00700
2	0.00000	0.01600	0.00000	0.09100	0.00000	0.16600	0.09000	1.00000	1.00000	0.01100
3	1.00000	0.01100	0.00000	0.09100	0.00000	0.22400	0.15700	1.00000	0.00000	0.01100
4	1.00000	0.01700	1.00000	0.09100	0.00000	0.36600	0.24700	0.00000	0.00000	0.01400
5	1.00000	0.00300	0.00000	0.39400	1.00000	0.21700	0.18000	1.00000	1.00000	0.01400

Figure 5 - following scaling, all values are constrained within the range of [0, 1].

Top of Form

5. Models

Our dataset is small and unbalanced. Therefore, to ensure that the various models we will examine have enough data to learn significant patterns, we chose to divide our data in a ratio of 80% train and 20% test.

5.1 Binary Logistic Regression

Since we are trying to predict the survival of a heart failure patient, the dependent variable distribution is Binomial, leading us to examine a generalized linear model (GLM). We will use binomial logit link function to model a binary outcome, which will give us a GLM called binomial logistic regression.

5.1.1 Model assumptions [13]

- **The dependent variable is binary** (death event is 1 for true or 0 for false)
- **Little or no multicollinearity** between explanatory variables - beforehand, we used VIF indicator to make sure the multicollinearity level will be under 10 (rule of thumb).
- **Linear relationship** - between the log odds and the different explanatory variables.
- **No extreme outliers** - we removed outliers within the interval of $3 \cdot IQR$ in the dataset preparation stage.
- **Independent observations–** each observation accounts for a different patient.
- **Sample size** - Logistic regression requires quite large sample sizes - Our data set is not large but not very small either so let's assume this assumption holds.

5.1.2 Model selection

Initially, we aim to explore potential interactions involving factorial variables to incorporate into our comprehensive model. After the EDA phase we would like to examine the following interactions:

1. **Anemia and ejection_fraction** (figure 6)

   Anemia may worsen the impact of low ejection fraction on the risk of death. Reduced oxygen delivery due to anaemia, combined with impaired heart function reflected by low ejection fraction, could synergistically increase the risk of adverse outcomes.

2. **Diabetes and serum_creatinine** (figure 7)

   In individuals with diabetes, high serum creatinine levels (indicative of kidney dysfunction) may pose a greater risk of death. The metabolic abnormalities associated with diabetes can exacerbate the adverse effects of impaired kidney function on cardiovascular health, leading to increased mortality risk.

3. **High_blood_pressure and ejection_fraction** (figure 8)

   High blood pressure and low ejection fraction may interact to elevate the risk of death. The combination of increased cardiac workload from hypertension and impaired cardiac function reflected by low ejection fraction can synergistically compromise cardiovascular health, contributing to higher mortality risk.

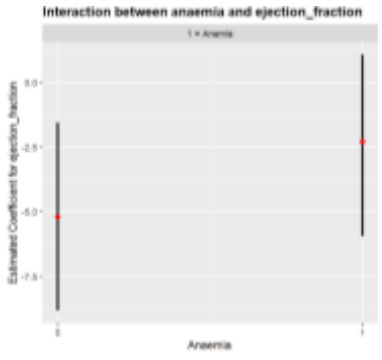


Figure 6- anemia & ejection fraction.

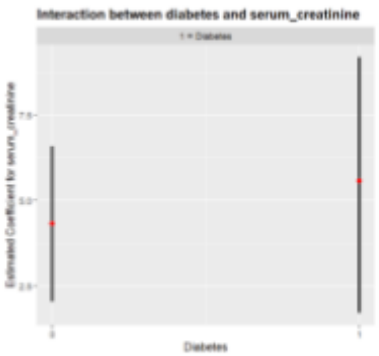


Figure 7 - diabetes & serum creatinine.

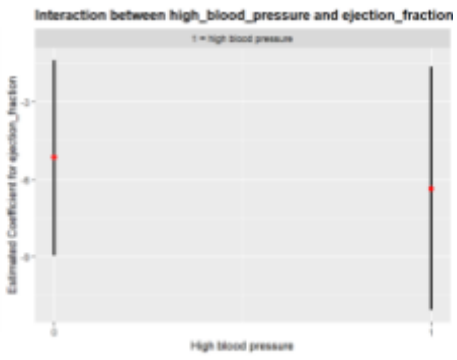


Figure 8 - high blood pressure & ejection fraction.

We utilized interplot [14] to stratify the data according to the categories of the factor variable (e.g., diabetes: yes/no) and visually illustrate the interaction effect within each category through individual panels. We observed **non-parallel lines** in the coefficient range, suggesting a potential interaction effect. This indicates that the impact of one health condition on the outcome could vary in strength depending on the level of the other condition.

Therefore, we chose to include the interesting interactions in the full model (Appendix C).

Afterwards, we used 'step' algorithms to find the best GLM model (Appendix D).

The summary of the models:

Table 6 - Summary of the models

Model	Model Equation	Deviance	AIC	BIC
Null	$\log log\,(odds) = -0.926$	248.09	250.09	253.428
Full	$\log log\,(odds) = 2.651 - 5.781 \cdot ejection_{fraction} - 1.404 \cdot anaemia(1) - 0.154 \cdot creatinine_{phosphokinase} + 4.433 \cdot s + 4.456 \cdot ejection_{fraction} * anaemia + 1.539 \cdot serum_{creatinine} * diabetes - 1.223 \cdot ejection_{fraction} * high_{blood\ pressure}$	144.591	170.159	216.884
Stepwise Forward	$\log log\,(odds) = 1.421 - 6.725 \cdot time + 4.758 \cdot serum_{creatinine} - 3.611 \cdot ejection_{fraction}$	**147.17**	**155.171**	**168.521**
Backwards	$\log log\,(odds) = 2.334 - 6.907 \cdot time + 4.8$	143.43	155.431	175.456

5.1.3 The Chosen Model

We selected the model generated through Stepwise and Forward selection due to its superior performance as indicated by the lowest AIC and BIC values, as demonstrated in Table 6. Furthermore, upon examination, we found that the significance level of the interaction between anemia and ejection fraction did not meet our predetermined threshold of 5%.



```
> chosen <-  foward_model
> summary(chosen)

Call:
glm(formula = DEATH_EVENT ~ time + serum_creatinine + ejection_fraction,
    family = binomial(link = "logit"), data = df_train)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        1.4207     0.6389   2.223  0.02619 *
time              -6.7259     1.1108  -6.055 1.41e-09 ***
serum_creatinine   4.7581     1.1227   4.238 2.25e-05 ***
ejection_fraction -3.6119     1.1907  -3.034  0.00242 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 248.09  on 207  degrees of freedom
Residual deviance: 147.17  on 204  degrees of freedom
AIC: 155.17
```

```
> # Equation of the probability (phi) as a function of the explanatory variables
> phi_equation <- paste("PHI =", round(exp(coefficients[1]), 4), "* exp(", round(coefficients[2], 4), "* ejection_fraction +", round(coefficients[3], 4), "*
serum_creatinine +", round(coefficients[4], 4), "* time) / (1 +", round(exp(coefficients[1]), 4), "* exp(", round(coefficients[2], 4), "* ejection_fraction
+", round(coefficients[2], 4), "* serum_creatinine +", round(coefficients[4], 4), "* time))")
> # Print the equation
> print(phi_equation)
[1] "PHI = 4.1388 * exp( -6.7259 * ejection_fraction + 4.7581 * serum_creatinine + -3.6119 * time) / (1 + 4.1388 * exp( -6.7259 * ejection_fraction + 4.7581
* serum_creatinine + -3.6119 * time))"
>
> # Equation of the log(odds)
> # Construct the log odds equation
> log_odds_equation <- paste("Log(Odds) =",
+                            round(coefficients[1], 4), "+",
+                            round(coefficients[2], 4), "* time +",
+                            round(coefficients[3], 4), "* serum_creatinine +",
+                            round(coefficients[4], 4), "* ejection_fraction")
>
> # Print the log odds equation
> print(log_odds_equation)
[1] "Log(Odds) = 1.4207 + -6.7259 * time + 4.7581 * serum_creatinine + -3.6119 * ejection_fraction"
```

1.421

- Explanatories: time, serum creatinine and ejection fraction.
- Intercept: When all explanatory variables are at their reference levels (i.e., zero), the predicted probability of death event, denoted as π, is calculated to be approximately 80.5% ($\pi = \frac{4.139 \cdot e^{(0)}}{1+4.139 \cdot e^{(0)}} = \frac{4.139}{5.139} = 0.805 \rightarrow 80.5\,\%,$). This indicates that in the hypothetical scenario where there is no observation time, serum creatinine, or ejection fraction, there is a high likelihood of a death event following a heart failure event. However, it's important to note that this predicted probability **lacks practical significance**, as it represents an extreme scenario that is unlikely to occur in real-world situations.
- Coefficients:

  Time: A negative coefficient indicates that as time increases (assuming all other variables are constant), the likelihood of a death event decreases. This suggests a decreasing trend in the probability of experiencing a death event following a heart failure event with longer observation periods.

  Serum Creatinine: With a positive coefficient, higher serum creatinine levels are associated with an increased likelihood of a death event after a heart failure event. This implies that elevated serum creatinine levels correspond to a higher probability of experiencing a death event.

  Ejection Fraction: Similarly, a negative coefficient suggests that lower ejection fraction values, indicative of poorer heart function, are associated with a higher likelihood of experiencing a death event following a heart failure event.

Conclusions:

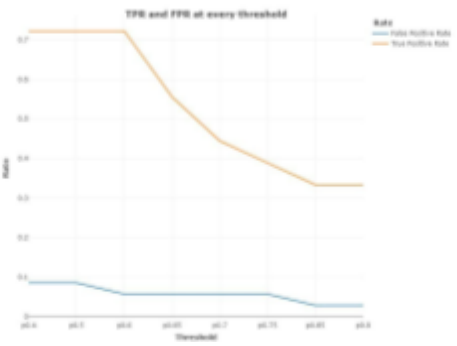
We infer that as the observation period increases and ejection fraction decreases (indicating poorer heart function), the likelihood of a death event decreases. Conversely, higher serum creatinine levels correspond to an increased likelihood of a death event following a heart failure event.

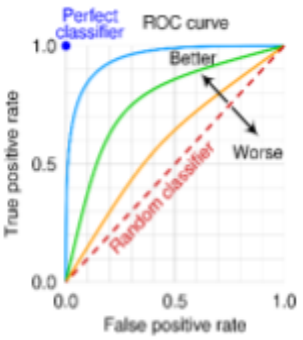
5.1.4 Tuning the threshold of the model

We wish to find the threshold that maximizes the orthogonal distance from the 45-degree line in the ROC curve that can be seen as maximizing the separation between the positive and negative classes. We aim to **find a point on the ROC curve that is closest to the top-left** corner, where **TPR is high**. This indicates that the model is correctly identifying a high proportion of true positives (death events).

For that, we calculated the Euclidean distance of each threshold i represented by $(FPR,\ TPR)_i$ to $(0, 1)$ and chose the threshold that minimized the distance. We examine the threshold to be within the values presented in the table below. Therefore, the selected threshold is **0.6.**

	Threshold	TPR	FPR	Distance
1	p0.4	0.7222222	0.08571429	0.2907016
2	p0.5	0.7222222	0.08571429	0.2907016
3	p0.6	0.7222222	0.05714286	0.2835944
4	p0.65	0.5555556	0.05714286	0.4481029
5	p0.7	0.4444444	0.05714286	0.5584866
6	p0.75	0.3888889	0.05714286	0.6137769
7	p0.85	0.3333333	0.02857143	0.6672786
8	p0.9	0.3333333	0.02857143	0.6672786

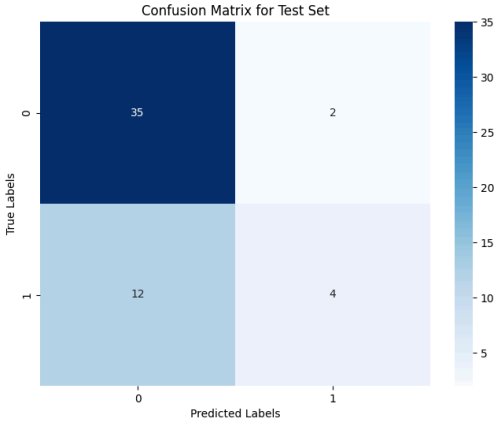




5.1.5 Model Scores

We applied a predetermined threshold of 0.6 to the test dataset, where any patient with a predicted probability of death equal to or exceeding 0.6 was classified as belonging to the DEATH_EVENT class. Upon comparing the predicted and true event classes, the model achieved the following scores:

Scoring method	Chosen model
AUC-ROC	0.907
F2 score	0.285
MCC	0.283



Correct distinctions - the model exhibits strong performance in accurately identifying cases where patients survive (**TN** = 35). However, it only predicts a small number of true positive cases where patients experience the "death_event" (**TP** = 4).

False distinctions - the model's inability to predict instances of the "death_event" (**FN** = 12) exceeds its incorrect predictions of patients experiencing the event when they survive (**FP** = 2).

5.2 Decision tree (DT)

A decision tree is a non-parametric supervised learning algorithm, which is utilized for both classification and regression tasks. It has a hierarchical, tree structure, which consists of a root node, branches, internal nodes and leaf nodes (Appendix E).

5.2.1 Model preparation

- Feature Independence: The Variance Inflation Factor (VIF) was calculated for each feature, helping to quantify the extent of multicollinearity and determine if features are independent enough for splitting nodes.
- No Missing Values: There are not missing values in the dataset.
- No Extreme Outliers: Outlier removal from the dataset was handled during the data preparation stage.

5.2.2 Hyperparameter Tuning

Hyperparameter tuning is the process of selecting the best set of hyperparameters for a machine learning model to optimize its performance. This action increases the accuracy score of the learning model to obtain an optimal model.

Using **GridSearchCV**, we aimed to find an optimized configuration for maximizing accuracy percentage by exploring potential ranges for each parameter. This process, conducted with cv=10, employed a consistent random_state value across all combinations of hyperparameter values to mitigate optimism. **We note that we performed the tuning with AUC-ROC scoring method uniformly in all the following models we tested.**

Here is the hyperparameter set:

- Criterion: The criterion hyperparameter determines the function used to measure the quality of a split. The two main options are Gini and Entropy. **Gini** measures the likelihood of an incorrect classification of a new instance if it were randomly classified according to the distribution of labels in the node. **Entropy** measures the amount of disorder or uncertainty in a set of data. The criterion that led to the highest accuracy is Entropy.
- Max depth: This hyperparameter controls the maximum depth of the tree. A high value can lead to overfitting. We chose to check the range between 1-10 (max depth of full tree). The Max depth that led to the highest accuracy is 3.
- Minimum leaf samples: This hyperparameter helps us control the growth of the tree and prevent overfitting. Setting a high value for this parameter ensures that each leaf has enough

samples to make reliable decisions. However, setting it too high can lead to underfitting, so we need to find a balance. Therefore, we started with relatively low values and gradually increased them while monitoring performance metrics. The range of values we checked was 1-10 (as the range is determined by the dataset size), and the maximum accuracy for the validation set was achieved with 7 leaf samples.

- Max features: This hyperparameter specifies the number of features to consider when looking for the best split. Using a smaller subset of features may prevent overfitting and speed up the training process. The range of values we checked was 1-10 (up to the number of features in our dataset), and the maximum accuracy for the validation set was achieved with 6 features.

5.2.3 Model Results

The Best hyperparameters summary:

Parameter	Default	After Tuning
Criterion	gini	entropy
Max depth	None	3
Minimum leaf samples	1	7
Max features	None	6
AUC-ROC on the test set	0.731	0.859
F2 score on the test set	0.625	0.759
MCC on the test set	0.462	0.681

5.3 Random forest (RF)

Random forests is an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time. For classification tasks, the output of the random forest is the class selected by most trees. For regression tasks, the mean or average prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set. Additionally, we made feature importance analysis to determine which variables contributed to the model the most (Appendix F).

5.3.1 Hyperparameter Tuning

We examined the same hyperparameters as those in the decision tree, including Max Features, Minimum Leaf Samples, Max Depth and Criterion. We decided to tune another hyperparameter which is unique for RF - **N_estimators**. This hyperparameter in a Random Forest determines the number of decision trees in the forest. Choosing a range of 50-300 trees (in duplicates of 50) is a common practice for dataset that is not large and is not particularly complex. This range helps to ensure that the random forest has enough trees to make stable predictions while trying to avoid overfitting.

5.3.2 Model Results

Parameter	Default	After Tuning
Criterion	gini	entropy
Max depth	None	5
Minimum leaf samples	1	2
Max features	Sqrt[4]	8
N estimators	100	50
AUC-ROC on the test set	0.866	0.872
F2 score on the test set	0.592	0.592
MCC on the test set	0.528	0.528

[4] square root of the total number of features



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5.4 Artificial Neural Network (ANN)

A neural network, drawing inspiration from the organization and functionality of the human brain, is a machine learning model comprising interconnected nodes known as neurons. These neurons process information and exchange signals through connections termed edges or weights. The primary objective of neural networks is to discern patterns within data, enabling them to make predictions or decisions based on these identified patterns. During training, neural networks adapt their weights by minimizing a loss function, which quantifies the disparity between their predictions and the actual outcomes (Appendix G).

5.4.1 Model preparation

- **Scaling -** we normalized data in both the training set and the test set, thereby ensuring consistency in the scale of features to avoid potential biases towards high-scale features in connections between neurons.
- **Missing values -** there are none missing values because we handle this in the pre-processing step.
- **No multicollinearity -** we incorporated the Variance Inflation Factor (VIF) during the pre-processing step to mitigate the issue of multicollinearity.

Examining the default network: We executed a neural network model using the default values. The number of neurons in the input layer is set to match the number of features in the dataset, which is 10. The single hidden layer contains 100 neurons.

5.4.2 Hyperparameter Tuning

- **Hidden layers -** indicates the number of hidden neuron layers and their size. Increasing the number of layers and neurons in each layer generally improves the model's fit to the training set, but many layers and neurons may result in overfitting. The range we chose to test was 1-3 layers and the number of neurons in each layer between 1 and 50 in increments of 5. We selected these ranges to capture more intricate patterns in the data. However, the range is not excessively large to prevent overfitting.
- **Activation function -** determines how the output of each neuron in the neural network is calculated and introduces nonlinearities into the model, allowing it to learn and represent complex relationships between the input features and the target variable. 'logistic', 'relu', 'tanh' and 'identity' are suitable for binary classification.

- **Learning rate -** defines how fast the network updates its parameters. A small value can result in very slow training, while a value that is too large can lead to rapid progress on the gradient and convergence to a suboptimal point. The range of values we chose to check is [0.00001, 0.0001, 0.001, 0.01, 0.1, 1, 10]. We selected this range by considering the default learning rate (0.001) and aimed to test several different orders of magnitude around it.
- **Maximum iterations -** the model will stop training when it reaches convergence or when it exceeds the maximum number of iterations. The range we tested [100, 200, 300, 400, 500], was chosen by assessing the default value of 200. This selection allows exploration of values both below and above the default to observe potential improvements in convergence or performance.

Presentation of the tested hyperparameter values is showcased as a correlation with accuracy percentages on the training and validation sets:

The chosen hyperparameters are 3 hidden layers with 11 neurons in each layer, learning rate of 0.001, maximum of 200 iterations, and an activation function of the 'tanh'.

Parameter	Default	Network training
Hidden layers sizes	(100)	(11, 11, 11)
Activation function	relu	tahn
Learning rate	0.001	0.001
Max iteration	200	200
AUC-ROC on the test set	0.9037	0.8969
F2 score on the test set	0.7051	0.6825
MCC on the test set	0.6314	0.6314

6. Discussion and Results

The objective of this project is to predict the survival of patients who have experienced heart failure based on their medical records. In our study, we assessed the performance of various models for this task: Generalized Linear Model (GLM), Decision Tree (DT), Random Forest (RF), and Artificial Neural Network (ANN).

To evaluate the **goodness of fit** of each model, we considered three indicators: AUC-ROC, Matthews Correlation Coefficient (MCC), and F2 score. These metrics are chosen to address the challenge of dealing with unbalanced data, which is prevalent in our dataset.

For a thorough assessment, we **equally weighed each indicator** and computed their average. This methodology enables us to pinpoint the index exhibiting the strongest performance, offering valuable insights for our project.

Summary of results:

Model /Score	AUC-ROC	MCC	F2 SCORE	Average Score
GLM	**0.907**	0.284	0.286	0.492
DT	0.859	**0.681**	**0.759**	**0.766**
RF	0.872	0.528	0.592	0.664
ANN	0.896	0.631	0.682	0.736

Based on the results presented in the table, we selected the Decision Tree (DT) model.

The analysis of the **DT model's confusion matrix** (Appendix E) reveals significant findings. The chosen model excels in predicting patient survival (True Negatives = 34) but falls short in predicting mortality (True Positives = 12). Despite a low false positive rate (False Positives = 3), it struggles with false negatives (False Negatives = 4). This highlights the model's strength in identifying survivors and the need for improvement in predicting mortality for better clinical reliability.

Our analysis has uncovered intriguing findings, emphasizing time, ejection fraction, and serum creatinine as the most **critical features** (Appendix E).

In conclusion, our project demonstrates the feasibility of utilizing an interpretable predictive model, such as Decision Trees (DT), to forecast patient mortality in cases of heart failure. However, it's essential to acknowledge several **limitations** encountered during the study: a relatively small dataset, imbalanced data distribution, and ambiguity surrounding the variable "time," which denotes the duration of treatment post-event. This lack of clarity raises questions regarding the treatment methodology, intervals between follow-ups, and related factors.



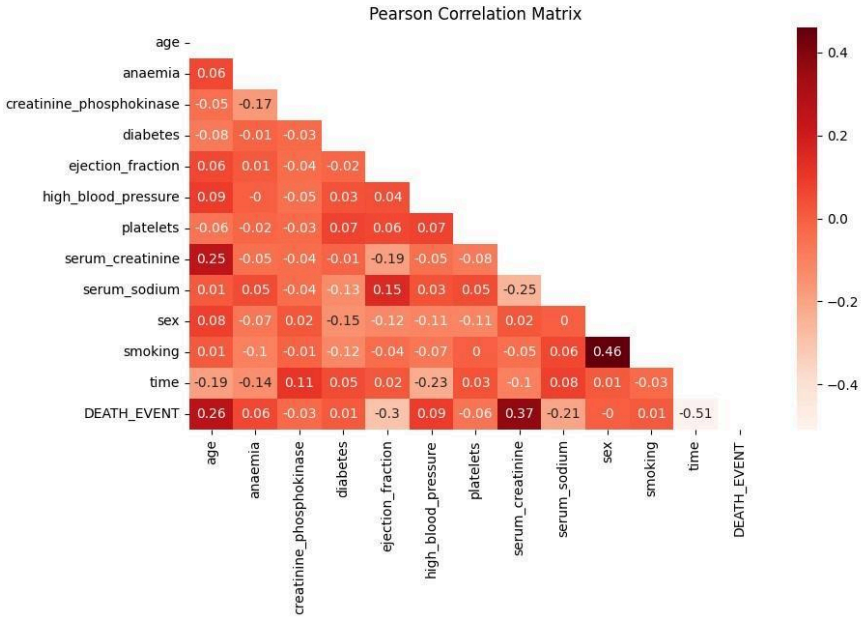
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8. Appendixes

Appendix A: correlation matrix after outlier's removal



The correlation matrix remains largely unchanged following the removal of outliers.

Appendix B: Categorical variables summary table

	Feature	Value	Number of Samples	Total (%)	Death events	Death (%)	Without death event	Without death (%)
0	anaemia	0	170	0.56856	50	0.16722	120	0.40134
1	anaemia	1	129	0.43144	46	0.15385	83	0.27759
2	diabetes	0	174	0.58194	56	0.18729	118	0.39465
3	diabetes	1	125	0.41806	40	0.13378	85	0.28428
4	high_blood_pressure	0	194	0.64883	57	0.19064	137	0.45819
5	high_blood_pressure	1	105	0.35117	39	0.13043	66	0.22074
6	sex	0	105	0.35117	34	0.11371	71	0.23746
7	sex	1	194	0.64883	62	0.20736	132	0.44147
8	smoking	0	203	0.67893	66	0.22074	137	0.45819
9	smoking	1	96	0.32107	30	0.10033	66	0.22074

Appendix C: Full and null models

Null model	Full model

Appendix D: Model selection using "Stepwise" algorithms

- **Stepwise**

```
> summary(step_model)

Call:
glm(formula = DEATH_EVENT ~ time + serum_creatinine + ejection_fraction,
    family = binomial(link = "logit"), data = df_train)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)         1.4207     0.6389   2.223  0.02619 *
time               -6.7259     1.1108  -6.055 1.41e-09 ***
serum_creatinine    4.7581     1.1227   4.238 2.25e-05 ***
ejection_fraction  -3.6119     1.1907  -3.034  0.00242 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 248.09  on 207  degrees of freedom
Residual deviance: 147.17  on 204  degrees of freedom
AIC: 155.17

Number of Fisher Scoring iterations: 6

> AIC(step_model)
[1] 155.171
> BIC(step_model)
[1] 168.5212
> formula(step_model)
DEATH_EVENT ~ time + serum_creatinine + ejection_fraction
```

```
> step_model <- step(null_model,
+                    scope = list(lower = null_model,
+                                 upper = full_model),
+                    direction = "both",
+                    data = df_train)
Start:  AIC=250.09
DEATH_EVENT ~ 1

                          Df Deviance    AIC
+ time                     1   186.45 190.45
+ serum_creatinine         1   225.25 229.25
+ ejection_fraction        1   227.20 231.20
<none>                         248.09 250.09
+ platelets                1   246.59 250.59
+ high_blood_pressure      1   247.34 251.34
+ anaemia                  1   247.92 251.92
+ creatinine_phosphokinase 1   247.95 251.95
+ sex                      1   248.01 252.01
+ smoking                  1   248.05 252.05
+ diabetes                 1   248.09 252.09

Step:  AIC=190.45
DEATH_EVENT ~ time

                          Df Deviance    AIC
+ serum_creatinine         1   157.53 163.53
+ ejection_fraction        1   168.94 174.94
<none>                         186.45 190.45
+ platelets                1   185.49 191.49
+ high_blood_pressure      1   185.74 191.74
+ smoking                  1   186.15 192.15
+ sex                      1   186.24 192.24
+ anaemia                  1   186.26 192.26
+ creatinine_phosphokinase 1   186.39 192.39
+ diabetes                 1   186.43 192.43
- time                     1   248.09 250.09

Step:  AIC=163.53
DEATH_EVENT ~ time + serum_creatinine

                          Df Deviance    AIC
+ ejection_fraction        1   147.17 155.17
<none>                         157.53 163.53
+ platelets                1   157.05 165.05
+ high_blood_pressure      1   157.20 165.20
+ creatinine_phosphokinase 1   157.36 165.36
+ sex                      1   157.43 165.43
+ anaemia                  1   157.44 165.44
+ smoking                  1   157.50 165.50
+ diabetes                 1   157.53 165.53
- serum_creatinine         1   186.45 190.45
- time                     1   225.25 229.25

Step:  AIC=155.17
DEATH_EVENT ~ time + serum_creatinine + ejection_fraction

                          Df Deviance    AIC
<none>                         147.17 155.17
+ sex                      1   146.84 156.84
+ high_blood_pressure      1   146.96 156.96
+ anaemia                  1   146.98 156.98
+ platelets                1   147.07 157.07
+ diabetes                 1   147.07 157.07
+ creatinine_phosphokinase 1   147.13 157.13
+ smoking                  1   147.15 157.15
- ejection_fraction        1   157.53 163.53
- serum_creatinine         1   168.94 174.94
- time                     1   211.54 217.54
```

- **Forward**

```
> summary(foward_model)

Call:
glm(formula = DEATH_EVENT ~ time + serum_creatinine + ejection_fraction,
    family = binomial(link = "logit"), data = df_train)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)        1.4207     0.6389   2.223  0.02619 *
time              -6.7259     1.1108  -6.055 1.41e-09 ***
serum_creatinine   4.7581     1.1227   4.238 2.25e-05 ***
ejection_fraction -3.6119     1.1907  -3.034  0.00242 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 248.09  on 207  degrees of freedom
Residual deviance: 147.17  on 204  degrees of freedom
AIC: 155.17

Number of Fisher Scoring iterations: 6

> AIC(foward_model)
[1] 155.171
> BIC(foward_model)
[1] 168.5212
> formula(foward_model)
DEATH_EVENT ~ time + serum_creatinine + ejection_fraction
```

```
> foward_model <- step(null_model,
+                      scope = list(lower = null_model,
+                                   upper = full_model),
+                      direction = "forward",
+                      data = df_train)
Start:  AIC=250.09
DEATH_EVENT ~ 1

                         Df Deviance    AIC
+ time                    1   186.45 190.45
+ serum_creatinine        1   225.25 229.25
+ ejection_fraction       1   227.20 231.20
<none>                        248.09 250.09
+ platelets               1   246.59 250.59
+ high_blood_pressure     1   247.34 251.34
+ anaemia                 1   247.92 251.92
+ creatinine_phosphokinase 1  247.95 251.95
+ sex                     1   248.01 252.01
+ smoking                 1   248.05 252.05
+ diabetes                1   248.09 252.09

Step:  AIC=190.45
DEATH_EVENT ~ time

                         Df Deviance    AIC
+ serum_creatinine        1   157.53 163.53
+ ejection_fraction       1   168.94 174.94
<none>                        186.45 190.45
+ platelets               1   185.49 191.49
+ high_blood_pressure     1   185.74 191.74
+ smoking                 1   186.15 192.15
+ sex                     1   186.24 192.24
+ anaemia                 1   186.26 192.26
+ creatinine_phosphokinase 1  186.39 192.39
+ diabetes                1   186.43 192.43

Step:  AIC=163.53
DEATH_EVENT ~ time + serum_creatinine

                         Df Deviance    AIC
+ ejection_fraction       1   147.17 155.17
<none>                        157.53 163.53
+ platelets               1   157.05 165.05
+ high_blood_pressure     1   157.20 165.20
+ creatinine_phosphokinase 1  157.36 165.36
+ sex                     1   157.43 165.43
+ anaemia                 1   157.44 165.44
+ smoking                 1   157.50 165.50
+ diabetes                1   157.53 165.53

Step:  AIC=155.17
DEATH_EVENT ~ time + serum_creatinine + ejection_fraction

                         Df Deviance    AIC
<none>                        147.17 155.17
+ sex                     1   146.84 156.84
+ high_blood_pressure     1   146.96 156.96
+ anaemia                 1   146.98 156.98
+ platelets               1   147.07 157.07
+ diabetes                1   147.07 157.07
+ creatinine_phosphokinase 1  147.13 157.13
+ smoking                 1   147.15 157.15
```

- **Backward**

```
> summary(back_model)

Call:
glm(formula = DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    time + ejection_fraction:anaemia, family = binomial(link = "logit"),
    data = df_train)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                2.3336     0.9318   2.504  0.01227 *
ejection_fraction         -6.4981     2.0889  -3.111  0.00187 **
anaemia                   -1.4254     0.9809  -1.453  0.14619
serum_creatinine           4.8063     1.1466   4.192 2.77e-05 ***
time                      -6.9068     1.1651  -5.928 3.06e-09 ***
ejection_fraction:anaemia  4.6578     2.5534   1.824  0.06813 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 248.09  on 207  degrees of freedom
Residual deviance: 143.43  on 202  degrees of freedom
AIC: 155.43

Number of Fisher Scoring iterations: 6

> AIC(back_model)
[1] 155.4311
> BIC(back_model)
[1] 175.4563
> formula(back_model)
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    time + ejection_fraction:anaemia
```

Algorithms suggested models

```
> back_model <- step(full_model,
+                    scope = ~1,
+                    direction = "backward",
+                    data = df_train)
Start:  AIC=170.16
DEATH_EVENT ~ ejection_fraction * anaemia + creatinine_phosphokinase +
    serum_creatinine * diabetes + ejection_fraction * high_blood_pressure +
    platelets + sex + smoking + time

                                      Df Deviance    AIC
- creatinine_phosphokinase             1   142.18 168.18
- smoking                              1   142.29 168.29
- platelets                            1   142.35 168.35
- ejection_fraction:high_blood_pressure 1   142.39 168.39
- sex                                  1   142.53 168.53
- serum_creatinine:diabetes            1   142.53 168.53
<none>                                     142.16 170.16
- ejection_fraction:anaemia            1   145.17 171.17
- time                                 1   200.22 226.22

Step:  AIC=168.18
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + high_blood_pressure + platelets + sex + smoking +
    time + ejection_fraction:anaemia + serum_creatinine:diabetes +
    ejection_fraction:high_blood_pressure

                                      Df Deviance    AIC
- smoking                              1   142.32 166.32
- platelets                            1   142.37 166.37
- ejection_fraction:high_blood_pressure 1   142.41 166.41
- serum_creatinine:diabetes            1   142.53 166.53
- sex                                  1   142.57 166.57
<none>                                     142.18 168.18
- ejection_fraction:anaemia            1   145.20 169.20
- time                                 1   200.47 224.47

Step:  AIC=166.32
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + high_blood_pressure + platelets + sex + time +
    ejection_fraction:anaemia + serum_creatinine:diabetes + ejection_fraction:high_blood_pressure

                                      Df Deviance    AIC
- ejection_fraction:high_blood_pressure 1   142.52 164.52
- platelets                            1   142.52 164.52
- sex                                  1   142.59 164.59
- serum_creatinine:diabetes            1   142.69 164.69
<none>                                     142.32 166.32
- ejection_fraction:anaemia            1   145.38 167.38
- time                                 1   201.12 223.12


Step:  AIC=164.52
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + high_blood_pressure + platelets + sex + time +
    ejection_fraction:anaemia + serum_creatinine:diabetes

                                Df Deviance    AIC
- high_blood_pressure            1   142.58 162.58
- platelets                      1   142.73 162.73
- sex                            1   142.84 162.84
- serum_creatinine:diabetes      1   142.91 162.91
<none>                               142.52 164.52
- ejection_fraction:anaemia      1   145.68 165.68
- time                           1   201.62 221.62

Step:  AIC=162.58
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + platelets + sex + time + ejection_fraction:anaemia +
    serum_creatinine:diabetes

                                Df Deviance    AIC
- platelets                      1   142.78 160.78
- sex                            1   142.88 160.88
- serum_creatinine:diabetes      1   143.03 161.03
<none>                               142.58 162.58
- ejection_fraction:anaemia      1   145.87 163.87
- time                           1   204.51 222.51

Step:  AIC=160.78
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + sex + time + ejection_fraction:anaemia + serum_creatinine:diabetes

                                Df Deviance    AIC
- sex                            1   143.02 159.02
- serum_creatinine:diabetes      1   143.17 159.17
<none>                               142.78 160.78
- ejection_fraction:anaemia      1   146.13 162.13
- time                           1   204.78 220.78

Step:  AIC=159.02
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + time + ejection_fraction:anaemia + serum_creatinine:diabetes

                                Df Deviance    AIC
- serum_creatinine:diabetes      1   143.33 157.33
<none>                               143.02 159.02
- ejection_fraction:anaemia      1   146.44 160.44
- time                           1   205.78 219.78

Step:  AIC=157.33
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    diabetes + time + ejection_fraction:anaemia

                                Df Deviance    AIC
- diabetes                       1   143.43 155.43
<none>                               143.33 157.33
- ejection_fraction:anaemia      1   146.89 158.89
- serum_creatinine               1   164.25 176.25
- time                           1   206.64 218.64


Step:  AIC=155.43
DEATH_EVENT ~ ejection_fraction + anaemia + serum_creatinine +
    time + ejection_fraction:anaemia

                                Df Deviance    AIC
<none>                               143.43 155.43
- ejection_fraction:anaemia      1   146.98 156.98
- serum_creatinine               1   164.39 174.39
- time                           1   206.64 216.64
```

Appendix E: Decision Tree

best parameters

```
Best hyperparameters: {'criterion': 'entropy', 'max_depth': 3, 'max_features': 6, 'min_samples_leaf': 7}
```

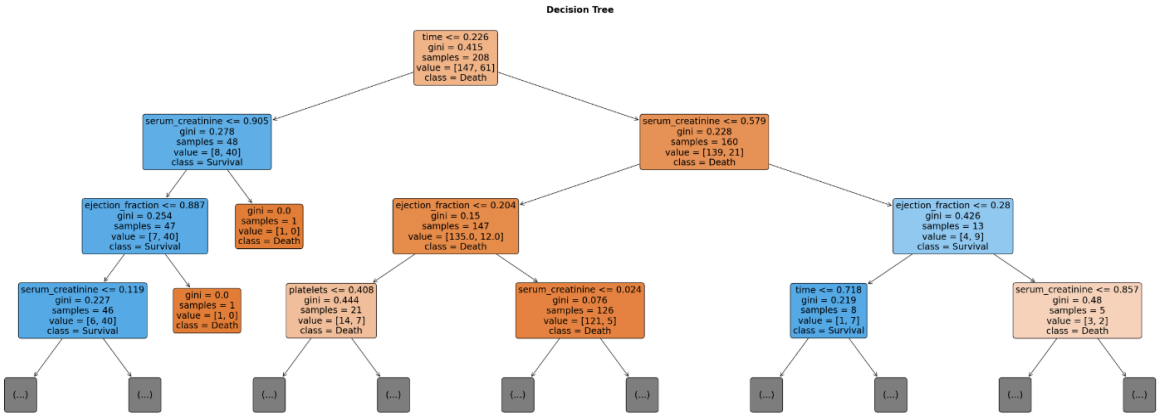
AUC-ROC score before tuning

```
AUC-ROC score for train set Decision Tree: 1.0
AUC-ROC Score for test set Decision Tree: 0.7314189189189189
```

AUC-ROC score after tuning

```
AUC-ROC Score for train set Decision Tree after tuning: 0.9490353518456562
AUC-ROC Score for test set Decision Tree after tuning: 0.8589527027027027
```

Best Decision Tree plot



MCC & F2 score (before tuning)

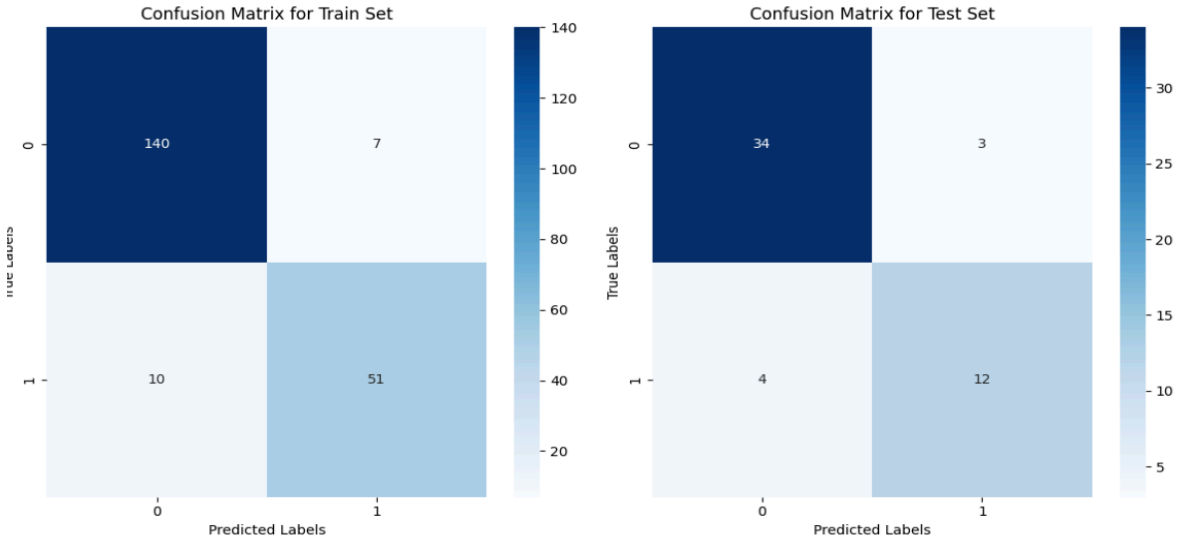
```
MCC for Decision Tree before tuning: 0.46283783783783783
F2 Score for Decision Tree before tuning: 0.625
```

F2 score (after tuning)

```
MCC for Decision Tree after tuning: 0.6817056676852529
F2 Score for Decision Tree after tuning: 0.759493670886076
```

Feature importance

Confusion matrix (tuned model)



Appendix F: Random Forest

best parameters

```
Best hyperparameters: {'criterion': 'entropy', 'max_depth': 5, 'max_features': 8, 'min_samples_leaf': 2, 'n_estimators': 50}
```

AUC-ROC score before tuning

```
AUC-ROC Score for train set Random Forest: 1.0
AUC-ROC Score for test set Random Forest: 0.8665540540540541
```

AUC-ROC score after tuning

```
AUC-ROC Score for train set Random Forest after tuning: 0.9957622393219583
AUC-ROC Score for test set Random Forest after tuning: 0.8716216216216216
```

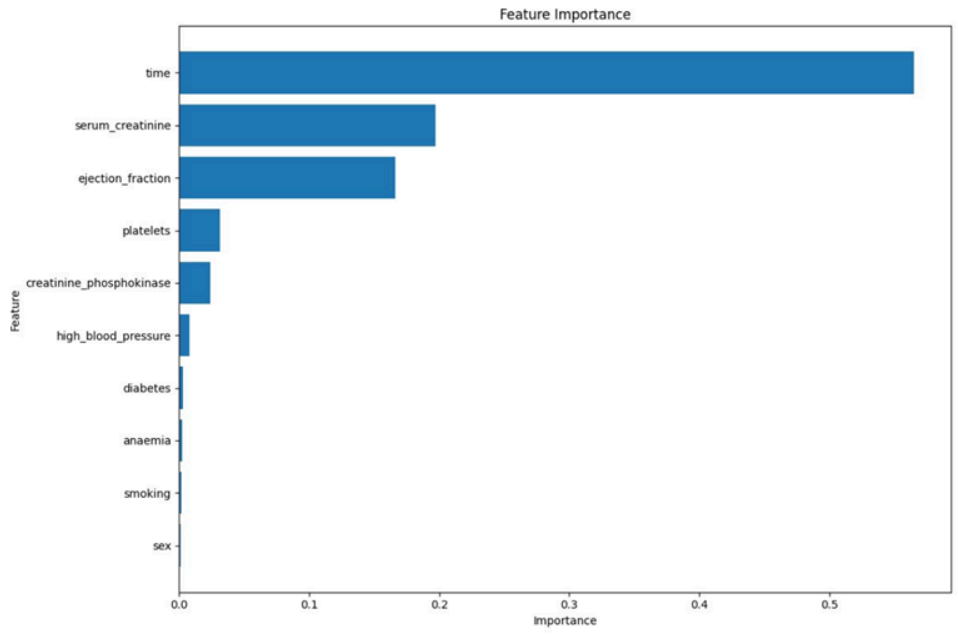
MCC & F2 score (before tuning)

```
MCC for Random Forest before tuning: 0.5280821103080681
F2 Score for Random Forest before tuning: 0.5921052631578947
```

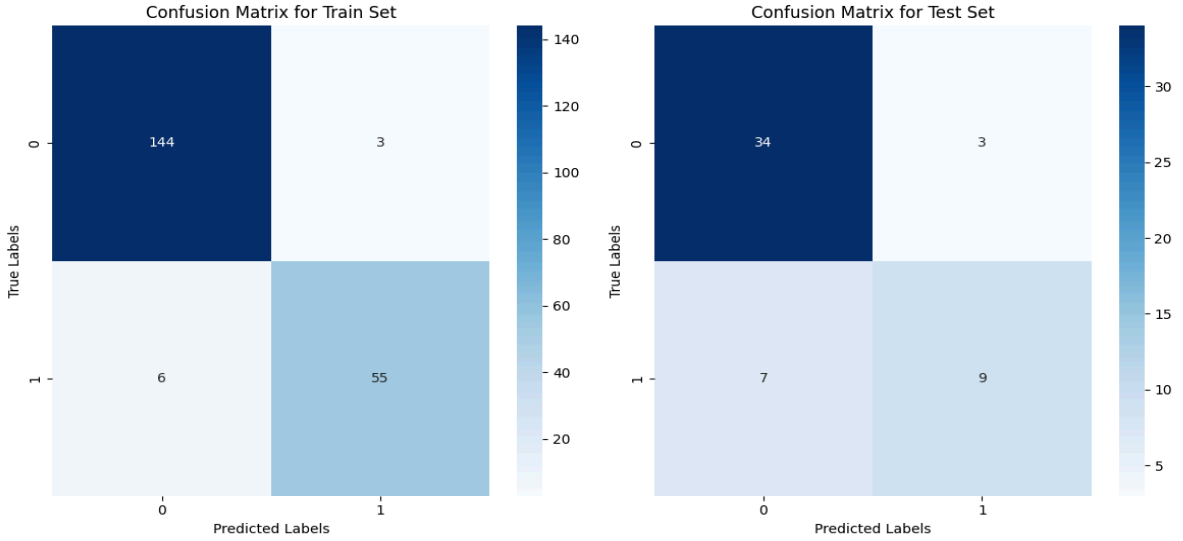
F2 score (after tuning)

```
MCC for Random Forest after tuning: 0.5280821103080681
F2 Score for Random Forest after tuning: 0.5921052631578947
```

Feature importance



Confusion matrix (tuned model)



Appendix G: ANN model

Default configuration

The default model configuration

```
activation : relu
alpha : 0.0001
batch_size : auto
beta_1 : 0.9
beta_2 : 0.999
early_stopping : False
epsilon : 1e-08
hidden_layer_sizes : (100,)
learning_rate : constant
learning_rate_init : 0.001
max_fun : 15000
max_iter : 200
momentum : 0.9
```

```
Number of neurons in the input layer: 10
Number of hidden layers: 1
Number of hidden neurons in each layer: (100,)
```

Default model ANN - Survival prediction results on the training set and the data set

```
AUC-ROC Score for MLP default on train set:  0.9106724657075945
AUC-ROC Score for MLP default on test set:  0.9037162162162161
```

```
F2 Score for MLP default using MinMax Scaler on train set:  0.6551724137931034
F2 Score for MLP default using MinMax Scaler on test set:  0.7051282051282052
```

```
MCC Score for MLP default using MinMax Scaler on train set:  0.623648415313423
MCC Score for MLP default using MinMax Scaler on test set:  0.6314475373900769
```

Adjusting parameters and choosing the best configuration

Hyperparameter Tuning

Hidden layers

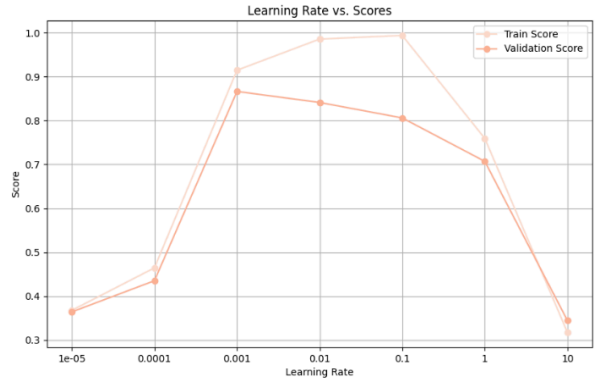
Hidden layers	Train Score	Validation Score
(21, 21)	0.928598	0.878844
(26, 26)	0.933569	0.877018
(41, 41)	0.948995	0.873696
(16, 16, 16)	0.933788	0.870522
26	0.907776	0.868435
(46, 46)	0.958616	0.862585
(11, 11, 11)	0.902038	0.862415
(31, 31)	0.926465	0.859342
(36, 36)	0.947302	0.857222
46	0.899551	0.857143
36	0.905743	0.857098
41	0.912087	0.856644
(26, 26, 26)	0.963015	0.855998
(21, 21, 21)	0.947849	0.851950

Activation function

Activation	Train Score	Validation Score
identity	0.901705	0.870816
tanh	0.901554	0.869546
relu	0.914604	0.866281
logistic	0.710148	0.637766

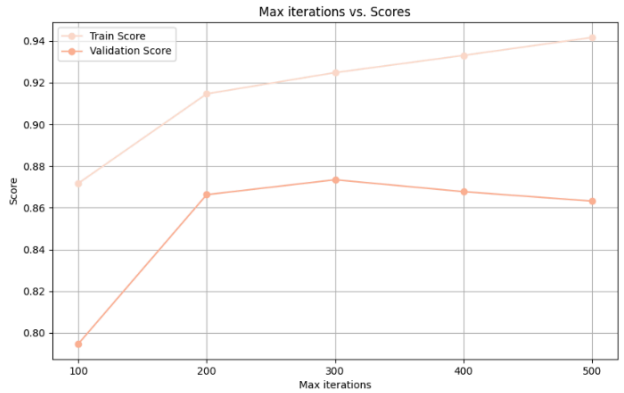
Learning rate

Learning rate	Train Score	Validation Score
0.001	0.914604	0.866281
0.01	0.985572	0.841179
0.1	0.993875	0.805930
1	0.759226	0.706944
0.0001	0.464196	0.434909
1e-05	0.367411	0.363889
10	0.317328	0.344365



Max iterations

Max iterations	Train Score	Validation Score
300	0.924782	0.873435
400	0.933028	0.867721
200	0.914604	0.866281
500	0.941635	0.863197
100	0.871702	0.794751



The best configuration selected

```
activation: tanh
alpha: 0.0001
batch_size: auto
beta_1: 0.9
beta_2: 0.999
early_stopping: False
epsilon: 1e-08
hidden_layer_sizes: (11, 11, 11)
learning_rate: constant
learning_rate_init: 0.001
max_fun: 15000
max_iter: 200
momentum: 0.9
n_iter_no_change: 10
nesterovs_momentum: True
power_t: 0.5
random_state: 42
shuffle: True
solver: adam
tol: 0.0001
validation_fraction: 0.1
verbose: False
warm_start: False
```

Confusion matrix

